

Gulam Sarvar

Associate Data Scientist · Voice AI · RAG Systems · LLM Applications
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PROFESSIONAL SUMMARY

Associate Data Scientist with production experience building **real-time voice AI platforms**, **RAG systems**, and **LLM applications**. Built **Kickcall**, a voice AI platform, from scratch — handling **150–200 daily calls** with **<2s end-to-end latency**. Led a cross-functional team of ~5 (frontend, cloud, backend) on production **RAG systems** across **legal** and **healthcare** client verticals. Author of **PyRAG Core**, an **open-source**, **Python-first** RAG platform supporting **pluggable vector backends** and **multi-provider LLM** integration.

TECHNICAL SKILLS

RAG & Retrieval: RAG Systems, Hybrid Retrieval, Reranking, Semantic/Hierarchical Chunking, LangChain, LangGraph, Multi-agent Workflows

Vector & Storage (Production): Qdrant, Pinecone, PostgreSQL, Redis

Vector & Storage (Project — PyRAG Core): Weaviate, Milvus, pgvector, Elasticsearch

Backend: Python, FastAPI, Pydantic, SQLAlchemy, Celery, Asyncio

LLM Providers: OpenAI, Anthropic, Gemini, OpenRouter, Ollama, vLLM, Hugging Face

Voice AI: WebRTC (LiveKit), STT/TTS Pipelines, Real-time Streaming Inference

Observability & DevOps: Langfuse, LangSmith, OpenTelemetry, Docker, Docker Compose

PROFESSIONAL EXPERIENCE

Associate Data Scientist

Apr 2024 – Present

Bluebash Pvt Ltd, Mohali

Kickcall — Voice AI Platform

- Built **Kickcall**, a production voice AI platform, from scratch — engineered the full **STT → LLM → TTS** streaming pipeline handling **150–200 calls/day** at **<2s end-to-end latency**, using **WebRTC (LiveKit)** for session management and **Redis** for state management across concurrent calls
- Designed **agent conversation flows**, **system prompts**, **fallback handling**, and turn-taking logic for inbound/outbound voice agents; optimised first-token latency through **parallel execution** and **streaming inference**
- Instrumented per-call, per-turn observability (**Langfuse**, **LangSmith**, **OpenTelemetry**) tracing LLM/STT/TTS response chunks, token usage, tool-call activation, and workflow state at each step — used for production debugging and latency optimisation
- Led initial **customer onboarding** post-sales handover — gathered business requirements and guided clients in configuring and customising voice agents for their specific use case

RAG Systems — Legal & Healthcare Clients

- Led a cross-functional team of ~5 (frontend, cloud, backend), assigning tasks per client requirements, building **RAG-based document intelligence systems** for two client verticals: **legal** (case-file and property-document analysis) and **healthcare** (patient intake form analysis), using **Qdrant/Pinecone hybrid retrieval** and **reranking** to reduce hallucinations on long-form documents
- Built **custom role-based admin dashboards** for both legal and healthcare clients, enabling each client's team to manage access, review workflows, and operate independently
- Integrated production observability (**Langfuse**, **LangSmith**, **OpenTelemetry**) to trace prompt failures and debug production regressions across both client systems

PROJECTS

PyRAG Core — Open-Source RAG Platform

GitHub

- Built an **open-source** RAG platform from scratch (**FastAPI**, **LangGraph**-native) with a pluggable vector-store abstraction — **Qdrant** is the production-proven path; four additional adapters (**Weaviate**, **Milvus**, **pgvector**, **Elasticsearch**) implement the same interface, pending live integration testing
- Implemented the full **ingestion pipeline** (parsing, configurable **chunking strategies**, **embedding**, indexing), **hybrid retrieval pipeline** (**dense + sparse retrieval**, **cross-encoder reranking**, context compression), and **multi-provider LLM** support: **OpenAI**, **Anthropic**, **Gemini**, **OpenRouter**, **Ollama**, **vLLM**
- Wrote **143 unit tests** covering ingestion, retrieval, all 6 LLM adapters (mocked), and vector store contracts; integrated **Langfuse** for cost, token, and latency tracking

EDUCATION

Master's in Computer Science (AI & ML)

2023 – 2026

Woolf University (Remote Program)

OPEN SOURCE

- Identified and reported a bug in **LiveKit Agents** (livekit/agents) encountered while building Kickcall's real-time session layer; acknowledged by maintainers and fixed upstream